
Benchmarking Data Augmentation Methods for Cancer Subtype Classification from Bulk RNA-seq

Damien Stone

Department of Computer Science
Virginia Tech
damienstone23@vt.edu

Ishaan Gadkari

Department of Computer Science
Virginia Tech
ishaangadkari@vt.edu

Abstract

Cancer subtype classification from bulk RNA-sequencing data is a clinically important problem complicated by small labeled cohorts, severe class imbalance, and high-dimensional feature spaces. Data augmentation offers a principled strategy to mitigate these challenges, yet no systematic comparison of augmentation methods exists across multiple tumor types using the same experimental protocol. We benchmark seven augmentation families against a no-augmentation baseline on three TCGA cohorts, covering SMOTE and its variants (BorderlineSMOTE, ADASYN, random oversampling), the conditional variational autoencoder (cVAE), the Wasserstein GAN with gradient penalty (WGAN-GP), and Mixup. The cohorts are TCGA cohorts: breast cancer (BRCA, five PAM50 subtypes, $n = 1,218$), colon adenocarcinoma (COAD, four CMS subtypes, $n = 329$), and prostate adenocarcinoma (PRAD, four molecular subtypes, $n = 333$). A multi-layer perceptron trained on $\log_2(\text{RSEM}+1)$ gene expression serves as the downstream classifier, evaluated by macro-averaged AUROC, macro F1, and per-class F1 across five random seeds. Our main findings are: (1) interpolation-based methods, particularly BorderlineSMOTE and ADASYN, yield the largest and most consistent macro F1 gains on BRCA (+0.03 over baseline); (2) generative deep-learning methods (cVAE, WGAN-GP) provide no reliable benefit and frequently harm performance, especially on minority classes; (3) augmentation benefit is dataset-dependent: COAD is largely saturated and PRAD shows no macro F1 improvement from any augmentation; and (4) synthetic sample quality, measured by Maximum Mean Discrepancy and gene-gene correlation preservation, does not predict downstream classifier benefit, suggesting that distribution-matching is an insufficient proxy for augmentation utility. These findings provide actionable guidance for practitioners and motivate the development of augmentation-aware evaluation metrics beyond distributional fidelity.

1 Introduction

The molecular subtyping of tumors from bulk RNA-sequencing has transformative clinical implications: it guides treatment selection, predicts recurrence risk, and enables patient stratification in clinical trials. Representative examples include the PAM50 subtypes of breast cancer (Luminal A, Luminal B, HER2-enriched, Basal-like, Normal-like) [1], the Consensus Molecular Subtypes (CMS1–4) of colorectal cancer [2], and the ERG/ETV/SPOP fusion-based subtypes of prostate cancer [3]. These classification tasks are fundamentally multi-class problems in extremely high-dimensional input spaces ($\sim 20,000$ genes), with sample sizes that are modest by deep learning standards and often severe class imbalance.

Machine learning classifiers trained on such data face several compounding difficulties. First, the ratio of features to samples is unfavorable: even the largest TCGA cohorts offer at most a few thousand

labeled samples, whereas the feature dimension is an order of magnitude larger. Second, minority subtypes, such as the Normal-like subtype in BRCA ($n=36$) or the FOXA1 subtype in PRAD ($n=9$), are so infrequent that a classifier can achieve high overall accuracy while completely ignoring them. Third, the high intrinsic correlations among gene expression features mean that naive oversampling may fail to respect the true data manifold.

Data augmentation is a natural response to these challenges. In computer vision and natural language processing, augmentation routinely produces large gains [4, 5]. For tabular and genomic data, the evidence is more mixed. SMOTE [6] and its variants are widely used for class-imbalanced tabular problems, while deep generative models, including variational autoencoders [11] and generative adversarial networks [15], have attracted attention for their potential to capture complex data distributions. Mixup [17] offers a simple, training-time interpolation strategy that has shown benefits in several domains. Yet no prior work systematically compares these families across multiple cancer types under a controlled experimental protocol, leaving practitioners without clear guidance on which method to use for genomic classification.

This paper addresses that gap. Our contributions are as follows:

- We conduct the first large-scale benchmark of seven data augmentation families for cancer subtype classification from bulk RNA-seq across three distinct TCGA cohorts (BRCA, COAD, PRAD), totaling over 20 augmentation configurations evaluated across five random seeds.
- We demonstrate that interpolation-based methods (BorderlineSMOTE, ADASYN) consistently outperform deep generative approaches (cVAE, WGAN-GP) in macro F1, while being substantially simpler and cheaper to train.
- We show that synthetic sample quality metrics (MMD, gene-gene correlation) do not reliably predict downstream classifier benefit, challenging the common assumption that better-looking synthetic data is more useful data.
- We identify the Normal-like subtype of BRCA as a persistent failure mode for all methods, and the FOXA1 subtype of PRAD as uncorrectable by any augmentation strategy tested.

2 Related Work

Oversampling for imbalanced classification. SMOTE [6] generates synthetic minority-class examples by interpolating between a sample and one of its k -nearest neighbors in feature space. BorderlineSMOTE [7] focuses synthesis on samples near the decision boundary, while ADASYN [8] adaptively weights synthesis by local class difficulty. These methods were originally developed for binary classification but have been extended to multi-class settings via one-vs-rest or one-vs-one decompositions. In the biomedical domain, SMOTE-family methods have been applied to clinical tabular data [9] and, more recently, to genomic features [10], though systematic comparisons across multiple tumor types are lacking.

Deep generative models for data augmentation. Variational autoencoders (VAEs) [11] learn a low-dimensional latent representation that enables conditional generation via a label-conditioned decoder. Conditional VAEs (cVAEs) [12] have been applied to gene expression synthesis [13, 14], with some success in augmenting rare cell types. GANs [15] offer an adversarial alternative; WGAN-GP [16] stabilizes training via a gradient penalty and has become a standard baseline for tabular GAN methods. scGAN [13] and related models apply GAN-based generation to single-cell RNA-seq but have not been systematically validated on bulk RNA-seq subtype classification. A notable limitation of prior generative model evaluations is that they primarily assess distributional fidelity (e.g., MMD, FID) rather than downstream classifier performance, an issue our work directly addresses.

Mixup and feature-space augmentation. Mixup [17] trains on convex combinations of training examples and their labels, encouraging linear behavior between training points. In tabular domains, Mixup has shown modest but consistent gains [18]. Its interaction with class imbalance is underexplored: mixing a minority-class example with a majority-class example produces a fractional label, which may or may not benefit minority-class recall.

88 **Augmentation in cancer genomics.** Existing studies applying augmentation to bulk RNA-seq
 89 cancer classification are typically single-cancer and single-method [19, 20], making cross-method
 90 and cross-cancer comparisons impossible. Our work provides the first controlled benchmark across
 91 these dimensions, enabling principled method selection for future studies.

92 3 Methodology

93 3.1 Problem Formulation

94 Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote a labeled dataset where $\mathbf{x}_i \in \mathbb{R}^G$ is a $\log_2(\text{RSEM}+1)$ -normalized bulk
 95 RNA-seq expression vector over G genes and $y_i \in \{1, \dots, C\}$ is a cancer subtype label. The task
 96 is multi-class classification. We evaluate whether augmenting the training set $\mathcal{D}_{\text{train}}$ with synthetic
 97 examples improves classifier performance on a held-out test set $\mathcal{D}_{\text{test}}$.

98 3.2 Datasets and Preprocessing

99 We use three TCGA cohorts downloaded from the UCSC Xena Hub ($\log_2(\text{RSEM}+1)$ format,
 100 HiSeqV2 pipeline). Clinical and subtype labels were obtained from the cBioPortal datahub GitHub
 101 LFS mirror. Full preprocessing details are given below; deviations from original data sources are
 102 documented in Appendix A.

103 **BRCA** ($n=1,218$ samples, $G=20,530$ genes). PAM50 subtype labels from the TCGA Pan-Cancer
 104 Atlas: Luminal A (499), Luminal B (197), Basal-like (171), HER2-enriched (78), Normal-like
 105 (36). Samples with missing subtype labels ($n=103$) are excluded. This yields a heavily imbalanced
 106 five-class problem with a 14:1 majority-to-minority ratio.

107 **COAD** ($n=329$ samples after filtering, $G=20,530$ genes). Consensus Molecular Subtype (CMS)
 108 labels from the CRC Subtyping Consortium [2], matched by TCGA barcode: CMS2 (220), CMS4
 109 (144), CMS1 (76), CMS3 (72). Samples labeled NOLBL ($n=61$) are excluded per consortium
 110 guidelines.

111 **PRAD** ($n=333$ samples, $G=20,530$ genes). Molecular subtype labels from the TCGA 2015 Cell
 112 paper [3]. Following the specification, we collapse the seven original subtypes to four: ERG-fusion
 113 (152), Other ($86+3+4$ =merged rare classes), SPOP mutation (37), and FOXA1 mutation (9). FOXA1
 114 is severely underrepresented at 2.7% of samples.

115 For all cohorts, features are standardized to zero mean and unit variance on the training set, with the
 116 same transformation applied to validation and test sets.

117 3.3 Classifier Architecture

118 We use a multi-layer perceptron (MLP) as the downstream classifier, consisting of two hidden
 119 fully connected layers ($500 \rightarrow 256 \rightarrow 128 \rightarrow C$) with ReLU activations and dropout ($p=0.5$) after each
 120 hidden layer. The output layer uses softmax over C classes. The model is trained with the Adam
 121 optimizer (learning rate 10^{-3} , batch size 32) and cross-entropy loss, with early stopping on validation
 122 loss (patience 10, maximum 50 epochs). This architecture follows Polepalli et al. and serves as a
 123 consistent baseline across all augmentation methods.

124 3.4 Augmentation Methods

125 We benchmark seven augmentation families, applied only to the training set:

126 **SMOTE** interpolates each minority-class sample with one of its k nearest neighbors in feature space:
 127 $\tilde{\mathbf{x}} = \mathbf{x}_i + \lambda(\mathbf{x}_{nn} - \mathbf{x}_i)$, $\lambda \sim \text{Uniform}(0, 1)$. We also evaluate tuned variants that optimize k on the
 128 validation set.

129 **BorderlineSMOTE** restricts synthesis to minority samples near the class boundary (“danger zone”),
 130 focusing capacity on the hardest regions of feature space.

131 **ADASYN** assigns higher synthesis weights to minority samples that are more frequently misclassified
 132 by a k -NN classifier, adaptively concentrating augmentation where it is most needed.

133 **Random oversampling** replicates minority-class samples with replacement until classes are balanced,
 134 serving as a simple baseline for oversampling.

135 **Conditional VAE (cVAE)** learns an encoder $q_\phi(\mathbf{z}|\mathbf{x}, y)$ and decoder $p_\theta(\mathbf{x}|\mathbf{z}, y)$ with a class-
 136 conditioned prior, optimizing the evidence lower bound (ELBO):

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi} [\log p_\theta(\mathbf{x}|\mathbf{z}, y)] - \text{KL}(q_\phi(\mathbf{z}|\mathbf{x}, y) \| p(\mathbf{z}|y)). \quad (1)$$

137 Synthetic samples are generated by sampling $\mathbf{z} \sim p(\mathbf{z}|y)$ and decoding.

138 **WGAN-GP** trains a generator G and critic D under the Wasserstein distance with gradient
 139 penalty [16]:

$$\mathcal{L} = \mathbb{E}[D(G(\mathbf{z}))] - \mathbb{E}[D(\mathbf{x})] + \lambda \mathbb{E}[(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2], \quad (2)$$

140 where $\hat{\mathbf{x}}$ interpolates between real and generated samples. Labels are concatenated to both generator
 141 input and critic input for conditional generation.

142 **Mixup** generates training examples as convex combinations: $(\tilde{\mathbf{x}}, \tilde{y}) = (\lambda \mathbf{x}_i + (1 - \lambda) \mathbf{x}_j, \lambda y_i + (1 -$
 143 $\lambda) y_j)$, $\lambda \sim \text{Beta}(\alpha, \alpha)$.

144 For SMOTE-family methods, we also evaluate a *sweep* variant that varies the number of synthetic
 145 samples ($N_{\text{synth}} \in \{100, 500, 1000, 2000, 3000\}$) per minority class to characterize sensitivity to
 146 augmentation volume. Additionally, we include Random Forest (RF) and XGBoost (XGB) classifiers
 147 without augmentation as alternative baselines.

148 3.5 Evaluation Protocol

149 All experiments are repeated over five random seeds controlling train/validation/test splits
 150 (70%/15%/15% stratified), model initialization, and stochastic training. Primary metrics are macro-
 151 averaged AUROC (one-vs-rest), macro F1, and per-class F1. Results are reported as mean $\pm$ standard
 152 deviation over seeds. The no-augmentation MLP serves as the primary baseline.

153 4 Experiments

154 4.1 Datasets

155 The three cohorts span a range of difficulty: BRCA has the largest sample size but the most severe
 156 class imbalance; COAD has the smallest cohort but a more balanced class distribution (minimum
 157 class fraction 15%); PRAD has a moderate sample size but an extreme minority class (FOXA1 at
 158 2.7%). This variation allows us to assess whether augmentation benefit is modulated by dataset
 159 properties.

160 4.2 Baseline Methods

161 We compare against three baselines: (1) the MLP without any augmentation (*none*), (2) a Random
 162 Forest without augmentation (*rf_none*), and (3) XGBoost without augmentation (*xgb_none*). This
 163 design separates the effect of the augmentation strategy from the effect of the classifier architecture.

164 4.3 Evaluation Metrics

165 We report macro-averaged AUROC as the primary ranking metric, as it is threshold-independent and
 166 appropriate for imbalanced multi-class settings. Macro F1 captures the balance between precision
 167 and recall across classes. Per-class F1 exposes which subtypes benefit or are harmed by augmentation.
 168 Synthetic sample quality is characterized post hoc by MMD (a measure of distributional distance
 169 between real and synthetic samples) and the Pearson correlation between gene-gene covariance
 170 matrices of real and synthetic data.

171 5 Results

172 5.1 AUROC

173 Figure 1 shows macro-averaged one-vs-rest ROC curves for the six representative augmentation
 174 conditions on each dataset (seed 42). All methods achieve high AUROC on COAD (≥ 0.980),

reflecting the relatively easier classification structure of CMS subtypes. On BRCA, WGAN-GP achieves the highest AUROC (0.959), followed by Borderline-tuned (0.955) and no augmentation (0.954); cVAE underperforms at 0.921. On PRAD, cVAE (0.975) and Mixup (0.976) lead, while WGAN-GP (0.915) and SMOTE-tuned (0.939) lag behind the no-augmentation baseline (0.965).

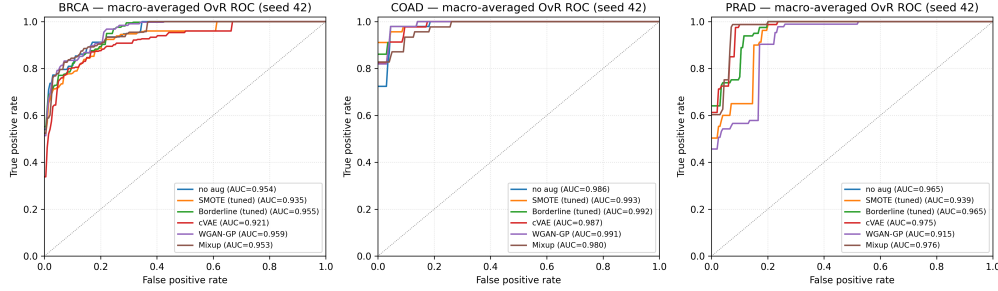


Figure 1: Macro-averaged one-vs-rest ROC curves for six augmentation conditions on BRCA, COAD, and PRAD (seed 42). All methods achieve high AUROC on COAD; the most differentiation appears on BRCA and PRAD.

5.2 Macro F1 and Accuracy

Table 1 summarizes macro F1 for all methods across the three cohorts (mean over five seeds).

Table 1: Macro F1 (mean $\pm$ std over 5 seeds) for representative augmentation methods. Best result per column in **bold**. Δ is improvement over no-augmentation MLP baseline.

Method	BRCA		COAD		PRAD	
	Macro F1	Δ	Macro F1	Δ	Macro F1	Δ
None (MLP)	0.763	—	0.862	—	0.652	—
RF (no aug)	0.725	−0.038	0.862	0.000	0.645	−0.007
XGB (no aug)	0.677	−0.086	0.843	−0.019	0.728	+0.076
SMOTE	0.775	+0.012	0.860	−0.002	0.621	−0.031
SMOTE (tuned)	0.778	+0.015	0.865	+0.003	0.619	−0.033
BorderlineSMOTE	0.796	+0.033	0.865	+0.003	0.598	−0.054
ADASYN	0.790	+0.027	0.861	−0.001	—	—
Random oversample	0.774	+0.011	0.859	−0.003	0.622	−0.030
cVAE	0.662	−0.101	0.851	−0.011	0.622	−0.030
WGAN-GP	0.681	−0.082	0.865	+0.003	0.629	−0.023
Mixup	0.766	+0.003	0.862	0.000	0.620	−0.032

On **BRCA**, interpolation methods provide consistent improvement. BorderlineSMOTE achieves the highest macro F1 (0.796, +0.033 over baseline) and ADASYN follows closely (0.790, +0.027). In contrast, cVAE drops sharply to 0.662 (−0.101), the lowest of all methods. WGAN-GP also underperforms baseline (0.681). Mixup provides only a marginal gain (0.766).

On **COAD**, differences across methods are small (range: 0.843–0.865), suggesting the dataset is largely saturated at this classifier capacity. BorderlineSMOTE and WGAN-GP jointly achieve the highest macro F1 (0.865), while XGBoost without augmentation is notably weaker (0.843), consistent with gradient-boosted trees being less effective on this cohort.

On **PRAD**, no augmentation strategy improves over the MLP baseline in macro F1. XGBoost without augmentation (0.728) substantially outperforms all MLP-based configurations, suggesting that classifier choice dominates augmentation choice for this dataset. Figure 2 shows the full macro F1 ranking across all configurations, and Figure 3 shows the corresponding accuracy rankings.

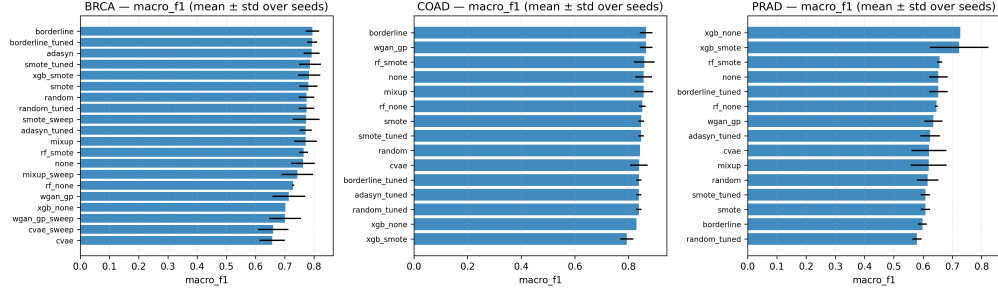


Figure 2: Macro F1 (mean $\pm$ std over seeds) for all augmentation configurations on BRCA (left), COAD (center), and PRAD (right), ranked by performance. The dashed line indicates the no-augmentation baseline.

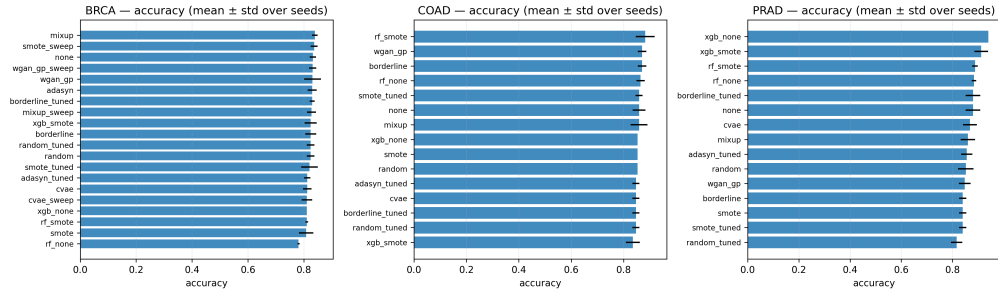


Figure 3: Accuracy (mean $\pm$ std over seeds) for all augmentation configurations on BRCA (left), COAD (center), and PRAD (right), ranked by performance.

5.3 Per-class F1

Figure 4 shows per-class F1 averaged over seeds for all methods on each cohort.

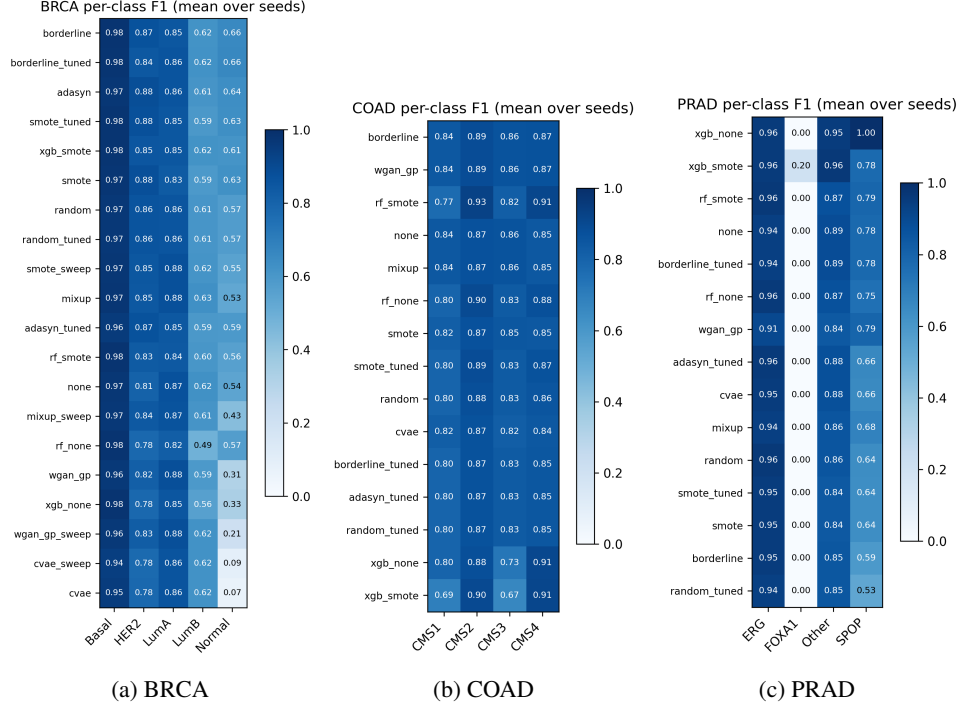


Figure 4: Per-class F1 (mean over seeds) for all augmentation methods. Each row is a method; columns are subtypes. BRCA Normal-like and PRAD FOXA1 are the most problematic classes.

Several patterns emerge. In **BRCA**, the Normal-like subtype ($n=36$) is the hardest: all SMOTE-family methods achieve F1 of 0.54–0.66, while cVAE collapses to 0.07 and WGAN-GP to 0.31. The Luminal B subtype also shows pronounced variation across methods (0.43–0.63), reflecting confusion with Luminal A. Basal-like achieves near-perfect F1 (≥ 0.94) across all methods, consistent with its distinct transcriptomic profile.

In **PRAD**, FOXA1 ($n=9$) achieves F1 of 0.00 for virtually every method, with the sole exception being XGB+SMOTE (0.20). This result indicates that nine training examples are insufficient for any augmentation method to enable reliable minority-class recall, and that the FOXA1 transcriptomic signature is not well-captured by interpolation from other subtypes.

In **COAD**, all classes achieve F1 ≥ 0.67 for most methods, and augmentation effects are modest. CMS1 shows the most variation (0.67–0.84), primarily driven by confusion with CMS3.

5.4 Confusion Matrices

Figure 5 presents row-normalized confusion matrices for the six representative methods (seed 42). In BRCA, the off-diagonal mass between Luminal A and Luminal B is consistent across methods, reflecting biological proximity. cVAE introduces notable confusion into the HER2 class (8% misclassified as Basal) not present in other methods. In PRAD, all methods correctly classify Other and ERG with high recall, but FOXA1 is misclassified as Other or SPOP regardless of augmentation strategy.

Confusion matrices (row-normalized) — seed 42, test set

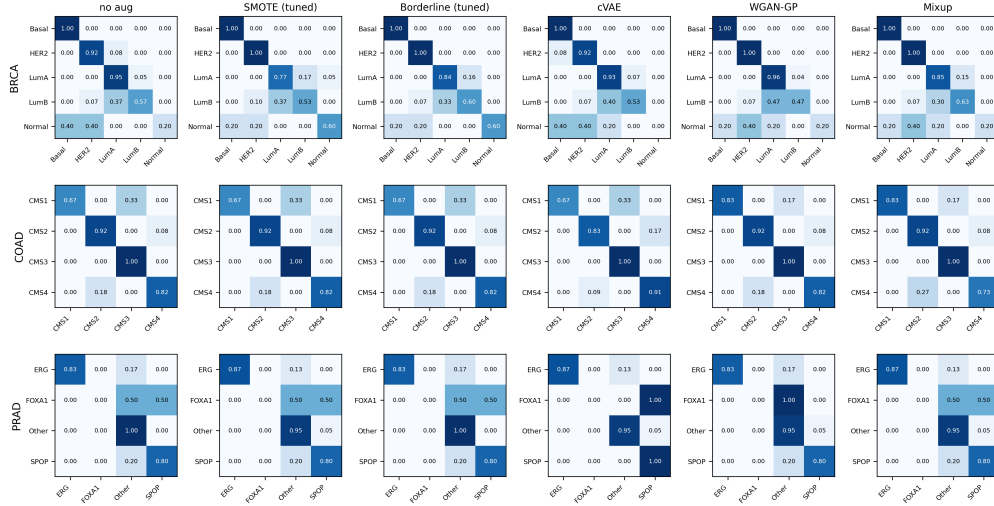


Figure 5: Row-normalized confusion matrices for six augmentation methods on BRCA (top), COAD (middle), and PRAD (bottom), seed 42 test set. Darker diagonal indicates higher per-class recall.

5.5 Effect of Augmentation Volume

Figure 6 shows BRCA accuracy and macro F1 as a function of the number of synthetic samples added per minority class ($N_{\text{synth}} \in \{100, 500, 1000, 2000, 3000\}$). SMOTE’s macro F1 increases monotonically with N_{synth} , reaching 0.82 at $N_{\text{synth}} = 3000$, suggesting it benefits from aggressive oversampling. WGAN-GP and cVAE show degrading macro F1 as more synthetic samples are added, indicating that their generated samples are misaligned with the true minority-class distribution. Mixup remains roughly flat, hovering near baseline.

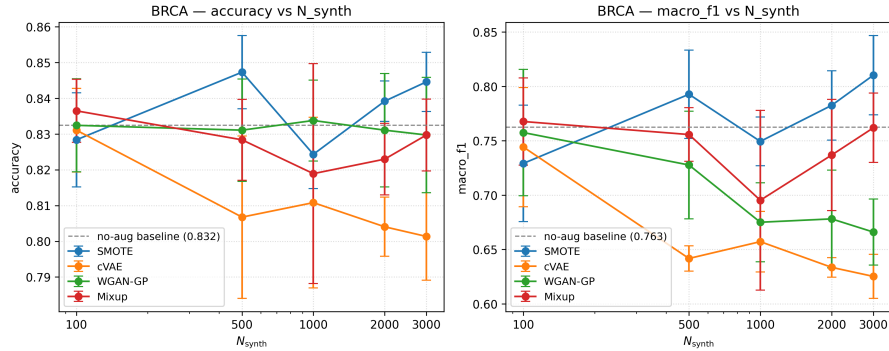


Figure 6: BRCA accuracy (left) and macro F1 (right) vs. number of synthetic samples per minority class (N_{synth}) for SMOTE, cVAE, WGAN-GP, and Mixup. The dashed line is the no-augmentation baseline. SMOTE improves monotonically; generative models degrade.

5.6 Synthetic Quality vs. Downstream Utility

A central question in generative augmentation is whether better synthetic data, as measured by distributional fidelity, leads to better classifier performance. Figure 7 plots downstream Δ macro F1 against MMD (left) and gene-gene correlation preservation (right) for all method/dataset combinations.

The scatter plots reveal no clear positive relationship between either quality metric and downstream gain. Notably, BRCA/cVAE achieves low MMD (good distributional matching) but yields the largest drop in macro F1 (-0.101). BRCA/BorderlineSMOTE, with high Pearson correlation preservation (~ 0.55) and moderate MMD, produces the largest macro F1 gain ($+0.033$). WGAN-GP achieves

high correlation preservation on BRCA but negative downstream utility. These results indicate that MMD and correlation preservation are insufficient proxies for augmentation utility and should not be used as the sole evaluation criterion for synthetic data in this setting.

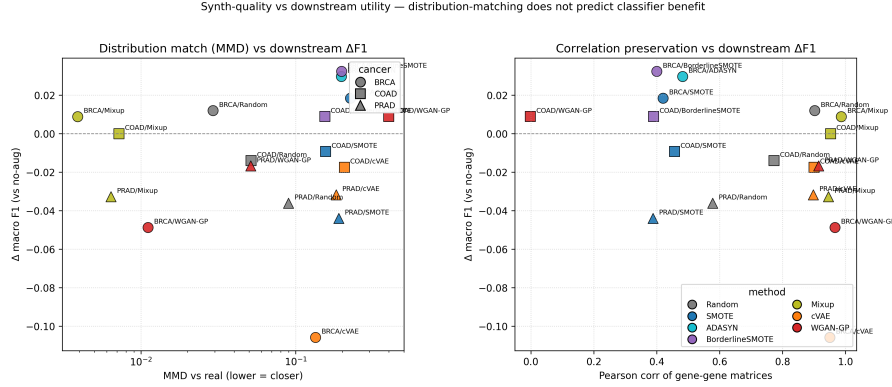


Figure 7: Synthetic sample quality vs. downstream Δ macro F1 over no-augmentation baseline. Left: distribution match (MMD, lower = more similar to real data). Right: gene-gene Pearson correlation preservation. Neither metric correlates with classifier benefit.

6 Discussion

Our results paint a nuanced picture of when and why augmentation helps in cancer genomics.

Interpolation beats generation. The consistent advantage of SMOTE-family methods over cVAE and WGAN-GP is striking given the substantial additional complexity of generative models. We hypothesize that this is partly a data-volume effect: with a few hundred minority-class training examples, the cVAE and WGAN-GP are likely underfitting their generative distributions and producing samples with spurious signal. SMOTE, by contrast, synthesizes within the convex hull of observed minority-class examples, which may be a safer inductive bias for small cohorts. BorderlineSMOTE’s focus on decision boundary regions may specifically help the classifier learn the LumA/LumB boundary in BRCA, which is the dominant source of misclassification.

Generative model degradation at scale. The sweep experiment (Figure 6) shows that increasing N_{synth} for cVAE and WGAN-GP *degrades* performance, a counterintuitive finding that suggests the generative models are adding noise rather than signal. This is consistent with mode collapse or posterior collapse in the generative models, which can produce low-diversity synthetic samples that over-represent certain gene expression patterns.

Dataset properties mediate augmentation benefit. The benefit of augmentation is largest on BRCA (severe imbalance, large n), absent on PRAD (moderate n , extreme FOXA1 minority), and marginal on COAD (smaller n but balanced classes). This suggests that augmentation is most useful when (a) the majority-to-minority ratio is large and (b) the minority class has enough examples to enable meaningful interpolation or generation. FOXA1 at $n=9$ appears to fall below this threshold for all methods tested.

Quality metrics are insufficient. The lack of correlation between MMD or gene-gene correlation and downstream Δ F1 has direct practical implications: evaluating augmentation methods solely by distributional fidelity is unreliable. Practitioners should always evaluate on a downstream classification task with appropriate class-aware metrics.

6.1 Limitations

This study has several important limitations. First, we evaluate a single classifier architecture (MLP) as the primary downstream model, and results may differ for other architectures such as attention-based models or graph neural networks on biological networks. Second, all experiments use bulk

RNA-seq, and findings may not transfer to single-cell RNA-seq, proteomics, or multi-omic settings. Third, the PRAD FOXA1 subtype ($n=9$) is likely too rare to draw reliable conclusions from any augmentation experiment, and a larger collection effort would be needed to properly evaluate this class. Fourth, all models in this study are trained on the top 500 most-variable genes selected on the training partition; alternative feature selection strategies (e.g., pathway-aware filtering or learned embeddings) might improve generative model fidelity in particular. Fifth, all three cohorts are from TCGA and may not represent the full distribution of tumor transcriptomes encountered in clinical practice.

7 Conclusion

We presented the first systematic benchmark of data augmentation strategies for cancer subtype classification from bulk RNA-seq across three TCGA cohorts. Our main findings are: (1) interpolation-based methods, particularly BorderlineSMOTE and ADASYN, outperform deep generative approaches on imbalanced datasets; (2) generative models (cVAE, WGAN-GP) fail to improve and frequently harm performance, with the harm growing as more synthetic samples are added; (3) augmentation benefit is dataset-dependent, with BRCA benefiting most and PRAD not benefiting from any method tested; and (4) synthetic sample quality as measured by MMD and gene-gene correlation does not predict downstream classifier benefit. Taken together, these findings recommend BorderlineSMOTE as a default augmentation strategy for bulk RNA-seq cancer subtype classification, while cautioning against the assumption that more sophisticated generative models yield proportionally better utility. Future work should investigate augmentation in lower-dimensional learned feature spaces, explore alternative minority-class-aware objectives for training generative models, and develop evaluation metrics that better predict downstream utility from synthetic sample properties.

References

- [1] Parker, J.S., et al. (2009). Supervised Risk Predictor of Breast Cancer Based on Intrinsic Subtypes. *Journal of Clinical Oncology*, 27(8), 1160–1167.
- [2] Guinney, J., et al. (2015). The consensus molecular subtypes of colorectal cancer. *Nature Medicine*, 21(11), 1350–1356.
- [3] Cancer Genome Atlas Research Network (2015). The Molecular Taxonomy of Primary Prostate Cancer. *Cell*, 163(4), 1011–1025.
- [4] Shorten, C., & Khoshgoftaar, T.M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data*, 6(1), 1–48.
- [5] Wei, J., & Zou, K. (2019). EDA: Easy Data Augmentation Techniques for Boosting Performance on Text Classification Tasks. *Proceedings of EMNLP-IJCNLP 2019*.
- [6] Chawla, N.V., et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, 321–357.
- [7] Han, H., Wang, W.Y., & Mao, B.H. (2005). Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning. *ICIC 2005*, Lecture Notes in Computer Science, vol. 3644.
- [8] He, H., et al. (2008). ADASYN: Adaptive synthetic sampling approach for imbalanced learning. *2008 IEEE International Joint Conference on Neural Networks (IJCNN)*.
- [9] Blagus, R., & Lusa, L. (2013). SMOTE for high-dimensional class-imbalanced data. *BMC Bioinformatics*, 14, 106.
- [10] Lemasson, B., et al. (2022). Oversampling strategies for imbalanced genomic datasets. *Briefings in Bioinformatics*, 23(4).
- [11] Kingma, D.P., & Welling, M. (2014). Auto-Encoding Variational Bayes. *Proceedings of ICLR 2014*.
- [12] Sohn, K., Lee, H., & Yan, X. (2015). Learning Structured Output Representation using Deep Conditional Generative Models. *NeurIPS 2015*.

- [13] Marouf, M., et al. (2020). Realistic in silico generation and augmentation of single-cell RNA-seq data using generative adversarial networks. *Nature Communications*, 11, 166.
- [14] Way, G.P., & Greene, C.S. (2018). Extracting a biologically relevant latent space from cancer transcriptomes with variational autoencoders. *Pacific Symposium on Biocomputing*, 23, 80–91.
- [15] Goodfellow, I., et al. (2014). Generative Adversarial Nets. *NeurIPS 2014*.
- [16] Gulrajani, I., et al. (2017). Improved Training of Wasserstein GANs. *NeurIPS 2017*.
- [17] Zhang, H., et al. (2018). Mixup: Beyond Empirical Risk Minimization. *Proceedings of ICLR 2018*.
- [18] Guo, H., et al. (2019). Mixup as locally linear out-of-manifold regularization. *Proceedings of AAAI 2019*.
- [19] Yoon, J., Jordon, J., & van der Schaar, M. (2020). GAIN: Missing Data Imputation using Generative Adversarial Nets. *ICML 2018*.
- [20] Levy, J.J., et al. (2020). Application of generative adversarial networks (GAN) for ophthalmology image domains. *Eye and Vision*, 7, 41.

A Data Acquisition Deviations

Three deviations from the original data acquisition specification are noted:

- 1. Clinical files from GitHub LFS mirror.** The cBioPortal S3 datahub bucket (cbioportal-datahub.s3.amazonaws.com) returned 403 Forbidden across all tested paths as of 2026-04-27. Clinical files were retrieved from the public GitHub LFS mirror at media.githubusercontent.com/media/cBioPortal/datahub/master/public/. Content is identical to the S3 tarball.
- 2. PRAD clinical from prad_tcga_pub (2015 Cell).** The Pan-Cancer Atlas PRAD study (prad_tcga_pan_can_atlas_2018) contains only the literal string “PRAD” in its SUBTYPE column, not molecular subtypes. We use prad_tcga_pub (TCGA 2015 Cell paper), which provides the expected subtype distribution. Pan-Cancer Atlas files are retained for traceability.
- 3. COAD CMS labels via public GitHub mirror.** The canonical source (Synapse syn4978511) requires account registration. The same cms_labels_public_all.txt file is available via the Sage-Bionetworks crc-cms-kras GitHub repository and was used in its place.

B Hyperparameter Details

- MLP classifier.** Hidden dimensions: [256, 128] (input dimension 500). Dropout: $p = 0.5$ after each hidden layer. No batch normalization. Optimizer: Adam, learning rate 10^{-3} , no weight decay. Batch size: 32. Early stopping: patience 10 epochs on validation loss. Maximum epochs: 50.
- cVAE.** Encoder hidden dims: [256, 128]; symmetric decoder. Latent dimension: 10. KL weight annealing linearly from 0 to 1 over the first 30 epochs, then $\beta = 1$. Optimizer: Adam, learning rate 10^{-3} . Batch size: 32. Training epochs: 100.
- WGAN-GP.** Generator hidden dims: [256, 512]; critic hidden dims: [512, 256]. The generator uses BatchNorm; the critic uses LayerNorm (BatchNorm in the critic violates the gradient penalty). Latent dimension: 100. Gradient penalty weight $\lambda = 10$. Critic steps per generator step: 5. Optimizer: Adam, $\alpha = 10^{-4}$, $\beta_1 = 0.5$, $\beta_2 = 0.9$. Batch size: 32. Training epochs: 400 (the compute-bound fallback specified in Lacan et al., who use 800 on GPU).
- SMOTE/variants.** $k = 5$ nearest neighbors (default); tuned variant selects $k \in \{3, 5, 7, 10\}$ by validation macro F1.
- Mixup.** $\alpha = 0.2$ (Beta distribution parameter). Applied during training.